Request Log — how this design came about

Reza Farivar’s instructions during the build session of 2026-10-20, quoted, with what each one produced. All of this was built in one session with Claude Code. The git history has the exact commits.

#	Reza's instruction (quoted)	What it produced
1	"help me build a package that generates knotted loop geometries with parameters that could shape the knotted loops … Ultimately, I want to build arrays of knotted loops, where each serves as a receiver loop in an MRI phased array. I want to be able to also 'bloom' the knots … Before you do it, tell me the design you understand here and then we'll freeze it in a Design.md file"	Design proposal: torus knots, bloom, arrays, exports, scope boundary (geometry only)
2	"by 'bloom into a loop' I mean with the topological overlap, not an actual loop."	Bloom collapses strands onto one loop-like footprint while keeping crossings
3	"can you also implement a feature that automatically calculates the optimal overlap between the coils for minimal mutual inductance, given the geometry? … our knots are pretty much flat, while respecting the topology"	Neumann-based automatic decoupling; flat knots with crossing lift
4	"I'd like to be able to put as many loops as I want, and the software decouples them given the bloom" · "Substrate is not PCB. We'll be using wire loops" · "I'd like to try up to 8 lobes" · "target array: helmet" · "new folder"	Unlimited N; wire loops; lobe catalogue 1–8; helmet former; `knotted_coils/`
5	"Freeze Design.md as v1.0 with the proposed defaults; I'll send wire and helmet specs separately. Start building."	Design v1.0 frozen (git 39a4e83, which pre-registers V1–V6); core, solver, validation, app
6	"we should be able to rotate the coils in-plane to help with decoupling. I think that should be one of the parameters"	Rotation exposed: free / locked / fixed
7	"Would I be able to visualize the e-fields in the UI via a toggle?"	Assessment: a quasi-static model gives only part of E. Deferred
8	"Make nearest-ring the default definition of decoupled (log it as a Design v1.1 change), and keep reporting the footprint verdict too."	Design v1.1. Disclosed that the worst-pair cost term had not been implemented
9	"Implement the worst-pair term as a smooth stand-in (soft-max or sum of k^4), compare against the current solver on the same seeds, and report both."	k^4 term; pre-registered 35-run paired comparison; adopted by rule; trade-off documented
10	"can I use the UI?" · "it seems frozen" · "why is it so clunky if I change a feature of the lobe?"	Local server; element checks moved to a worker; hidden-canvas freeze and slider bug fixed
11	"currently, 'bloom' is just pinching a portion of the loop. I want 'bloom' to make the lobes more extended, such that each lobe's aspect ratio would increase as I increase 'bloom'."	Design v1.2: lobes extend outward from a fixed base circle
12	"Add a petal-sharpness control that narrows each lobe, with the same topology checks."	Design v1.3 (power law). V7 FAILED → v1.4 (angle warp)
13	"can you plot the decoupling graph in the 'Pair explorer' tab in dB?"	dB view, $20 \cdot \log_{10}\lvert k \rvert$, labelled "not bench S21"

#	Reza's instruction (quoted)	What it produced
14	"is there a fast and effective way to visualize the e-fields of a solved array?" · "how do I visualize the B1 field?" · "I just want a quick visualization that's reasonable. We will do simulations later"	Design v1.5: quick B1 view (Biot–Savart; head surface and slices). E-fields left to full-wave simulation
15	"package the whole thing so I can share it" · "include all the instructions you gathered here (the design files, etc.) so that others can expand on the work"	This package: developer guide, request log, project CLAUDE.md, integrity rules, sources

Standing instructions that shaped the work

These come from Reza's workspace instructions (his global `CLAUDE.md`). The ones relevant to this project:

- Render a styled `.html`, and a `.pdf` where possible, alongside every Markdown file.
- Define jargon on first use; Reza's background is vision science and neuroscience, not software or engineering.
- Follow the research-integrity framework (copied in `docs/research_integrity/`):
  - never fabricate;
  - cite only verified sources;
  - tag parameter origins;
  - pre-register checks and report failures;
  - label unvalidated results exploratory;
  - attribute AI-written code.